Dual-Constrained Deep Reinforcement Learning for Joint UAV Trajectory Design and NOMA-Based Task Offloading in MEC Networks

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***Abstract*—Unmanned aerial vehicle (UAV)-assisted mobile edge computing (MEC) with non-orthogonal multiple access (NOMA) offers a promising paradigm for low-latency IoT task offloading in dynamic environments. However, jointly optimizing UAV trajectory, NOMA power allocation, and offloading decisions under hard deadline constraints is NP-hard and difficult to solve with conventional methods. In this paper, we formulate the problem as a constrained Markov decision process (CMDP) and propose DC-PPO (Dual-Constrained Proximal Policy Optimization), which incorporates a Lagrangian dual variable with a physically-derived upper bound $\lambda_{\max}$ to enforce delay constraints without destabilizing the on-policy training dynamics. The key insight of DC-PPO is a reward-cost ratio upper bound $\lambda_{\text{crit}} = \bar{r}/\bar{c}$, which ensures the augmented reward remains positive throughout training. Extensive experiments with $N{=}20$ users demonstrate that DC-PPO achieves a deadline satisfaction rate of 82.5%, outperforming unconstrained PPO by 7.3 percentage points ($\mathbf{p = 0.031}$, Cohen's $d = 2.69$) and reducing training variance by 47% compared to PPO-Lagrangian. Scalability experiments confirm that the performance advantage grows monotonically with user density, reaching $+6.8$ pp at $N{=}25$. An unexpected finding is the *NOMA interference saturation effect*: myopic full-offloading achieves only 15.4% deadline satisfaction—far below random (63.0%)—because simultaneous maximum-power transmission collapses the NOMA SINR for all users.**

***Index Terms*—UAV-MEC, NOMA, task offloading, constrained Markov decision process, proximal policy optimization, Lagrangian relaxation, deadline constraint.**

I. Introduction

THE proliferation of compute-intensive IoT applications—ranging from real-time video analytics to autonomous vehicle coordination—is straining the processing capacity of resource-constrained edge devices. Mobile Edge Computing (MEC) addresses this by deploying computation servers at the network edge, enabling low-latency task offloading [1], [2]. However, deploying MEC infrastructure at fixed base stations is impractical in dynamic or disaster-affected environments where ground connectivity is disrupted. Unmanned aerial vehicles (UAVs) equipped with MEC servers offer a compelling complement: their mobility allows on-demand repositioning to serve user clusters, and line-of-sight (LoS) channels to ground devices yield significantly higher transmission rates than terrestrial links [3], [4].

UAV-assisted MEC faces a fundamental throughput bottleneck when serving multiple users simultaneously: traditional orthogonal multiple access (OMA) schemes allocate disjoint frequency bands, limiting spectral efficiency as the number of users grows. Non-orthogonal multiple access (NOMA), which superposes multiple users on the same channel and relies on successive interference cancellation (SIC) at the receiver, can serve N users simultaneously within the same bandwidth [5]–[7], making it a natural fit for the dense IoT offloading regime.

Challenges and Motivation.

Jointly optimizing *UAV trajectory*, *NOMA power allocation*, and *computation offloading ratio* presents three inter-coupled challenges:

C1 – High-dimensional continuous action space. Even for $N{=}20$ users, the joint action space has 42 continuous dimensions (UAV displacement + per-user offloading ratio + per-user NOMA power), ruling out discrete-action RL methods such as DQN and making tabular or convex approaches intractable.

C2 – Hard deadline constraints. IoT tasks carry strict latency deadlines (e.g., $t_{\max}{=}1.0\,\text{s}$ for compressed image analytics), yet the delay depends non-linearly on the NOMA interference structure, which varies with UAV position and user power levels. Existing unconstrained RL methods cannot reliably enforce such constraints during training.

C3 – Lagrangian instability. The Lagrangian relaxation approach [8] converts the constrained optimization into a penalized objective, but the dual multiplier λ_d must be carefully bounded. Without a principled upper bound, λ_d can grow without limit, causing the augmented reward to become negative and collapsing the policy gradient—a failure mode we term *Lagrangian saturation*.

Related Work.

UAV trajectory optimization. Conventional approaches formulate UAV-MEC as a mixed-integer non-linear program and apply successive convex approximation (SCA) or block coordinate descent [3], [9]–[11]. These methods yield locally optimal solutions but require accurate channel state information and cannot handle stochastic task arrivals without recomputation each slot.

Deep RL for MEC offloading. DRL methods learn offloading policies directly from environment interactions, avoiding the need for explicit channel modeling [12], [13]. DDPG and TD3 handle continuous action spaces but require large replay buffers and struggle in high-dimensional settings under limited training budgets [14]. On-policy methods such as PPO offer more stable training but lack a principled mechanism for constraint enforcement. DC-PPO uses identical hyperparameters and network architecture to its PPO baseline (128 hidden,

Tanh, orthogonal init, obs normalization, entropy coeff. 0.01, clip ϵ=0.2); the sole difference is the Lagrangian mechanism.

Constrained RL. Reward Constrained Policy Optimization (RCPO) [8] and Constrained Policy Optimization (CPO) [15] provide theoretical frameworks for CMDP-based learning. CPO relies on a computationally expensive trust-region projection that scales poorly to 42-dimensional action spaces. RCPO's Lagrangian approach is more practical, but it does not specify how to bound the multiplier, leading to the saturation problem identified in this work. More recent methods include FOCOPS [16], which constrains the first-order policy update, and CUP [17], which uses a constrained update projection. However, these methods require computing the constraint cost gradient, which adds significant overhead in high-dimensional continuous action spaces. PID-Lagrangian methods [18] apply proportional-integral-derivative control to the multiplier update to reduce oscillation, but still lack a principled upper bound. To our knowledge, no prior work has derived a *physically-grounded* upper bound for the Lagrangian multiplier in the UAV-MEC context.

UAV-NOMA-MEC. Recent works have combined UAV trajectory design with NOMA offloading. Wang *et al.* [19] applied DRL to multi-UAV MEC task scheduling, and Zhang *et al.* [20] studied energy-efficient UAV-NOMA for IoT with imperfect CSI. Chen *et al.* [21] addressed task-dependent delay in multi-UAV MEC scheduling. Hu *et al.* [22] jointly optimized UAV trajectory and offloading ratio for NOMA-MEC networks via SCA, demonstrating throughput gains over OMA baselines. Li *et al.* [23] applied DRL to NOMA-assisted UAV-MEC resource allocation but used an unconstrained reward without deadline enforcement. In contrast, DC-PPO explicitly handles deadline constraints via a physically-grounded Lagrangian bound, and provides statistical validation across 5 seeds with a 42-dimensional joint action space—a setting not addressed by prior work.

Contributions.

This paper proposes **DC-PPO** (Dual-Constrained PPO), which addresses the above challenges through three contributions:

- **C1 [CMDP Formulation]:** We provide an integrated CMDP formulation that jointly couples UAV trajectory, per-user NOMA power allocation, and partial offloading ratio in a single 42-dimensional continuous action space, with an episode-level delay constraint explicitly capturing the NOMA interference structure under SIC decoding.
- **C2 [$\lambda_{\max}$ Selection Criterion]:** We derive a *reward-cost ratio upper bound* $\lambda_{\text{crit}} = \bar{r}/\bar{c}$ that guarantees the Lagrangian-augmented reward remains positive throughout training (Lemma 2). Setting $\lambda_{\max}$ to a physically-derived fraction of λ_{crit} prevents *Lagrangian saturation* (the collapse of the policy gradient when λ_d exceeds λ_{crit}), reducing training variance by 47% compared to unconstrained PPO-Lagrangian.
- **C3 [Empirical Validation]:** Extensive experiments with N=20 users and 5 independent seeds demonstrate that DC-PPO achieves 82.5% deadline satisfaction, outperforming unconstrained PPO by 7.3 percentage points (p=0.031, Cohen's d=2.69). Scalability experiments confirm that the DC-PPO advantage grows monotonically with user density, reaching +6.8 pp at N=25.

The remainder of this paper is organized as follows. Section II presents the system model and CMDP formulation. Section III describes DC-PPO, including the $\lambda_{\max}$ derivation and algorithm pseudocode. Section IV details the experimental setup. Section V presents and analyzes the results. Section VI concludes the paper.

II. System Model and Problem Formulation

A. Network Topology

We consider a UAV-assisted MEC system operating over T discrete time slots $\mathcal{T} = \{1, \ldots, T\}$, each of duration $\Delta t = 1$ s. A single UAV equipped with an MEC server serves N ground IoT users $\mathcal{N} = \{1, \ldots, N\}$ distributed uniformly at random in a square area of side $L = 200$ m. The UAV flies at a fixed altitude $H = 100$ m and communicates with users via NOMA uplink on a bandwidth of $B = 5$ MHz. Let $\boldsymbol{q}_t = (x_t, y_t)^\top \in [0, L]^2$ denote the UAV's horizontal position at slot t.

B. UAV Mobility Model

The UAV moves horizontally between consecutive slots. The action $\boldsymbol{a}_t^{(\text{fly})} = (\delta x_t, \delta y_t)^\top$ specifies the desired displacement, subject to a maximum speed constraint:

$$\|\boldsymbol{a}_t^{(\text{fly})}\|_2 \le v_{\max} \cdot \Delta t, \tag{1}$$

with $v_{\max} = 20$ m/s. The UAV position is updated as:

$$\boldsymbol{q}_{t+1} = \text{clip}\Big(\boldsymbol{q}_t + \boldsymbol{a}_t^{(\text{fly})},\ [0, L]^2\Big). \tag{2}$$

The propulsion energy consumed per slot is modeled as [3]:

$$E_t^{(\text{fly})} = \big(c_1 + c_2 v_t^2\big)\,\Delta t, \tag{3}$$

where $v_t = \|\boldsymbol{a}_t^{(\text{fly})}\|_2/\Delta t$ is the flight speed, $c_1 = 80$ W is the hovering power, and $c_2 = 0.12\,\text{W}\cdot\text{s}^2/\text{m}^2$ is a velocity-dependent coefficient.

C. Air-to-Ground NOMA Channel

1) Channel Model: The air-to-ground channel from user i to the UAV follows a probabilistic LoS model [3]. Let $d_i = \|\boldsymbol{q}_t - \boldsymbol{u}_i\|_2$ be the horizontal distance, where $\boldsymbol{u}_i \in \mathbb{R}^2$ is user i's fixed location. The LoS probability is:

$$P_{\text{LoS},i} = \frac{1}{1 + a\exp(-b(\theta_i - a))}, \tag{4}$$

where $\theta_i = \arctan(H/d_i)$ (in degrees) is the elevation angle, and $a = 9.61$, $b = 0.16$ are environment-specific parameters for urban areas [3]. The average path loss (linear) is:

$$\text{PL}_i = \text{PL}_i^{\text{fs}} \cdot \left(P_{\text{LoS},i} \cdot \eta_{\text{LoS}} + (1 - P_{\text{LoS},i}) \cdot \eta_{\text{NLoS}}\right), \tag{5}$$

where $\text{PL}_i^{\text{fs}} = (4\pi f_c \ell_i / c)^2$ is the free-space path loss with $\ell_i = \sqrt{d_i^2 + H^2}$, $f_c = 2$ GHz, and $\eta_{\text{LoS}} = 1$ dB, $\eta_{\text{NLoS}} = 20$ dB are the additional attenuation factors for LoS and NLoS links, respectively (linear equivalents: $10^{0.1} \approx 1.26$ and $10^2 = 100$). The channel power gain is $h_i = 1/\text{PL}_i$.

2) NOMA Uplink with SIC: In each slot, active users transmit simultaneously on the shared 5 MHz channel under NOMA. Let $p_i \in [0, p_{\max}]$ be the transmit power of user i, with $p_{\max} = 0.2$ W. The UAV receiver applies Successive Interference Cancellation (SIC). Let $\pi(1), \pi(2), \ldots, \pi(N)$ denote the SIC decoding order sorted by *descending* received power, i.e., $p_{\pi(1)}h_{\pi(1)} \geq p_{\pi(2)}h_{\pi(2)} \geq \cdots$. User $\pi(k)$ is decoded k-th, treating users $\pi(k{+}1), \ldots, \pi(N)$ as interference (users already decoded are perfectly cancelled). The SINR of the k-th decoded user is:

$$\text{SINR}_{\pi(k)} = \frac{p_{\pi(k)}h_{\pi(k)}}{\sum_{m=k+1}^{N} p_{\pi(m)}h_{\pi(m)} + \sigma^2}, \tag{6}$$

where $\sigma^2 = N_0 B = -110$ dBm is the total noise power over the $B{=}5$ MHz bandwidth, with noise power spectral density $N_0 = -174$ dBm/Hz. The achievable uplink rate is:

$$R_i = B \log_2(1 + \text{SINR}_i) \quad \text{(bits/s)}. \tag{7}$$

D. Task Model and Computation Offloading

1) Task Arrival: In each slot t, user i generates a computation task independently with probability $p_a = 0.5$. A task is parameterized by its data size $D_i \sim \mathcal{U}[0.4, 2.0]$ Mbit and computation density $\kappa_i \sim \mathcal{U}[500, 1000]$ cycles/bit, so the required computation is $C_i = D_i\kappa_i$ cycles.

2) Partial Offloading: For each active user i, the agent decides an offloading ratio $\alpha_i \in [0, 1]$ (fraction of task bits offloaded to UAV-MEC) and an NOMA transmit power $p_i \in [0, p_{\max}]$. The offloaded fraction $\alpha_i D_i$ is transmitted to the UAV at rate R_i and executed on the MEC server at an allocated CPU frequency $f_i^{(\text{MEC})}$. The remaining fraction $(1-\alpha_i)D_i$ is processed locally at CPU frequency $f_{\text{loc}} = 1$ GHz.

The end-to-end task completion delay for user i (denoted τ_i to distinguish from the horizontal distance d_i) is:

$$\tau_i = \max\left(\frac{(1-\alpha_i)C_i}{f_{\text{loc}}},\ \frac{\alpha_i D_i}{R_i} + \frac{\alpha_i C_i}{f_i^{(\text{MEC})}} \right), \tag{8}$$

where the $\max$ reflects parallel execution of local and offload portions (a standard assumption in partial offloading models [24]). We neglect MEC queuing delay, which is reasonable under the equal-frequency partitioning assumption. We adopt *equal frequency partitioning* as a tractable approximation that focuses the optimization on trajectory and offloading decisions: $f_i^{(\text{MEC})} = f_{\text{UAV}}/N_{\text{active},t}$, where $f_{\text{UAV}} = 8$ GHz and $N_{\text{active},t}$ is the number of active users in slot t. We note that variable frequency allocation could be included as an additional action dimension at the cost of a larger action space; we leave this extension for future work.

The user-side energy consumption is:

$$e_i = \kappa_{\text{e}} f_{\text{loc}}^2 \cdot (1-\alpha_i)C_i + p_i \cdot \frac{\alpha_i D_i}{R_i}, \tag{9}$$

where the first term is the local computation energy with effective capacitance coefficient $\kappa_{\text{e}} = 10^{-27}\,\text{J}\cdot\text{s}^2/\text{cycle}^2$ (so that $\kappa_{\text{e}} f_{\text{loc}}^2 C$ has units of Joules), and the second term is the transmission energy. A task is deemed to meet its deadline if $\tau_i \leq t_{\max} = 1.0$ s.

E. Problem Formulation as CMDP

The joint optimization of UAV trajectory, NOMA power allocation, and offloading ratio is formulated as a Constrained Markov Decision Process (CMDP) [25].

1) State Space: The state at slot t is:

$$\boldsymbol{s}_t = \left[\frac{\boldsymbol{q}_t}{L},\ \{d_i/L,\ \mathbb{1}[i \text{ active}],\ D_i/D_{\max},\ C_i/C_{\max},\ h_i'\}_{i=1}^{N},\ \frac{T-t}{T} \right] \in \mathbb{R}^{3+4N}, \tag{10}$$

yielding a 103-dimensional state for $N = 20$, where d_i is the horizontal UAV-to-user distance, $h_i' = \log_{10}(h_i + 10^{-12})$ is the log-normalized channel gain, and $D_{\max}$, $C_{\max}$ are normalization constants. User positions $\boldsymbol{u}_i$ are fixed within an episode and re-sampled uniformly at each episode reset, so the agent must generalize over different user distributions.

2) Action Space: The continuous action at each slot is:

$$\boldsymbol{a}_t = \left[\delta x_t,\ \delta y_t,\ \{\alpha_{i,t}\}_{i=1}^{N},\ \{p_{i,t}\}_{i=1}^{N} \right] \in \mathbb{R}^{2+2N}, \tag{11}$$

a 42-dimensional continuous space for $N = 20$. All components are mapped to $[-1, 1]$ via the neural network output and linearly scaled to their physical ranges.

3) Reward Function: The slot-level reward balances task throughput against UAV propulsion energy:

$$r_t = \sum_{i \in \mathcal{A}_t} w_i - \beta E_t^{(\text{fly})}/E_{\max}, \tag{12}$$

where $\mathcal{A}_t$ is the set of active users in slot t, $w_i = 1$ rewards the agent for each task handled (irrespective of deadline status, following the CMDP paradigm in which the objective and constraint are separated), $\beta = 0.01$ is a weighting coefficient that keeps the flight energy penalty small relative to task throughput, and $E_{\max} = (c_1 + c_2 v_{\max}^2)\Delta t = 128$ J is the maximum single-slot flight energy (at full speed), used to normalize the energy term to $[0, 1]$. Deadline satisfaction is enforced exclusively through the Lagrangian constraint in (15).

4) Constraint: We impose a soft delay constraint via a Lagrangian penalty. Let $\bar{\tau}_t \triangleq |\mathcal{A}_t|^{-1} \sum_{i \in \mathcal{A}_t} \tau_{i,t}$ denote the per-slot average task delay. The episode-level average delay constraint is:

$$\mathbb{E}_\pi \left[\frac{1}{T} \sum_{t=1}^{T} \bar{\tau}_t \right] \leq \tau^*, \tag{13}$$

where $\tau^* = 0.5$ s/task is the delay budget. The hard deadline is $t_{\max} = 1.0$ s; the soft budget $\tau^* = 0.5$ s provides an anticipatory signal that guides the policy to avoid deadline violations before they occur.

5) CMDP Objective: The optimization problem is:

$$\max_{\pi} \quad \mathbb{E}_\pi \left[\sum_{t=1}^{T} \gamma^{t-1} r_t \right] \tag{14}$$

$$\text{s.t.} \quad \mathbb{E}_\pi \left[\frac{1}{T} \sum_{t=1}^{T} \bar{\tau}_t \right] \leq \tau^*, \quad \text{(average delay budget)} \tag{15}$$

$$(1),\ (2),\ \alpha_{i,t} \in [0, 1],\ p_{i,t} \in [0, p_{\max}],$$

where π denotes the policy and $\gamma = 0.99$ is the discount factor. The CMDP (14)–(15) is computationally intractable in general.

Lemma 1 (NP-Hardness of Single-Slot Subproblem)**.** *The single-slot subproblem of* (14) *(fixed UAV position, equal power p_i=$p_{\max}$, binary offloading $\alpha_i\in\{0,1\}$) is NP-hard.*

Proof. We reduce from the *Maximum Feasible Subsystem* (MaxFS) problem, which is NP-hard [26]. With equal power p_i=$p_{\max}$ and binary α_i, the rate R_i of user i (if offloading) depends on the SIC decoding order of the offloading subset $\mathcal{S} = \{i : \alpha_i{=}1\}$. User $i \in \mathcal{S}$ meets its deadline iff: $D_i/R_i(\mathcal{S}) + C_i/f_i^{(\text{MEC})} \leq t_{\max}$, where $R_i(\mathcal{S})$ is a non-monotone function of $\mathcal{S}$ through SINR coupling (6). Finding the largest $\mathcal{S}$ such that all users in $\mathcal{S}$ meet their deadlines is equivalent to MaxFS over the SINR-rate-delay inequality system, which is NP-hard for $N \geq 3$ [26]. The continuous relaxations ($\alpha_i \in [0,1]$, variable p_i) do not reduce the complexity class. □

Since the single-slot subproblem is NP-hard and the full CMDP additionally optimizes over UAV trajectory (coupling channel gains h_i across slots), the multi-slot problem (14)–(15) is at least as hard, motivating the deep reinforcement learning approach proposed in Section III.

III. Proposed DC-PPO Algorithm

A. Lagrangian Relaxation of the CMDP

We solve the CMDP (14)–(15) via the Lagrangian dual method [8], [25]. Introducing a non-negative dual variable (Lagrangian multiplier) $\lambda_d \geq 0$ for the delay constraint (15), the primal problem is converted into an unconstrained penalized objective:

$$\mathcal{L}(\pi,\lambda_d) \triangleq \mathbb{E}_\pi\left[\sum_{t=1}^{T}\gamma^{t-1}r_t\right] - \lambda_d{\cdot}\mathbb{E}_\pi\left[\frac{1}{T}\sum_{t=1}^{T}\bar{\tau}_t - \tau^*\right]. \quad (16)$$

At each time step the agent observes the augmented reward:

$$r'_t = r_t - \lambda_d \cdot c_t, \quad c_t = \bar{\tau}_t, \quad (17)$$

where $c_t = \bar{\tau}_t$ is the per-task average delay in slot t. The multiplier λ_d is updated via projected gradient ascent after each episode:

$$\lambda_d \leftarrow \Pi_{[0,\,\lambda_{\max}]}(\lambda_d + \eta_\lambda\,(\bar{\tau}_{\text{ep}} - \tau^*))\,, \quad (18)$$

where $\bar{\tau}_{\text{ep}}$ is the episode-averaged per-task delay, $\eta_\lambda = 3\times 10^{-2}$ is the dual learning rate, and $\Pi_{[\cdot]}$ denotes projection onto $[0,\lambda_{\max}]$.

B. Physical Upper Bound for $\lambda_{\max}$

The choice of $\lambda_{\max}$ is the key design parameter of DC-PPO. Setting $\lambda_{\max}$ too small renders the constraint inactive; setting it too large causes the augmented reward r'_t to become negative, collapsing the policy gradient direction.

Lemma 2 (Policy Gradient Collapse under Lagrangian Saturation)**.** *Let $\bar{r} \triangleq \mathbb{E}_\pi[r_t]$ and $\bar{c} \triangleq \mathbb{E}_\pi[c_t]$ denote the steady-state mean reward and cost under a feasible policy π. Define the* critical multiplier *$\lambda_{crit} \triangleq \bar{r}/\bar{c}$. If $\lambda_d > \lambda_{crit}$, then:*

(i) *$\mathbb{E}_\pi[r'_t] < 0$, i.e., every action yields negative expected augmented return;*

(ii) *the policy gradient magnitude satisfies $\|\nabla_\theta J'(\theta)\| \leq \frac{2\gamma T}{(1-\gamma)^2}(\sigma_r + \lambda_d\sigma_c)\,\|\nabla_\theta \log\pi_\theta\|_\infty$, where $\sigma_r^2 = Var[r_t]$ and $\sigma_c^2 = Var[c_t]$. As $\lambda_d \to \infty$, the gradient direction is dominated by the cost term, and* all actions are penalized uniformly regardless of task completion quality.

Proof. **Part (i).** Taking expectations of the augmented reward (17): $\mathbb{E}[r'_t] = \bar{r} - \lambda_d\bar{c} < 0$ iff $\lambda_d > \bar{r}/\bar{c} = \lambda_{\text{crit}}$.

Part (ii). The augmented return is $J'(\theta) = \mathbb{E}_\pi[\sum_{t=1}^{T}\gamma^{t-1}r'_t]$. By the policy gradient theorem [27], $\nabla_\theta J' = \mathbb{E}_\pi[\sum_t \gamma^t Q'(s_t,a_t)\nabla_\theta\log\pi_\theta(a_t|s_t)]$ where $Q'(s,a) = \mathbb{E}[\sum_{l=0}^{\infty}\gamma^l r'_{t+l}|s_t{=}s, a_t{=}a]$. Since $|r'_t| \leq |r_t| + \lambda_d|c_t|$, we have $|Q'(s,a)| \leq \frac{1}{1-\gamma}(\bar{r} + \lambda_d\bar{c} + \sigma_r + \lambda_d\sigma_c)$. When $\lambda_d \gg \lambda_{\text{crit}}$, the $\lambda_d\sigma_c$ term dominates, and $\nabla_\theta J'$ aligns with $-\nabla_\theta\mathbb{E}[c_t]$ irrespective of the reward structure, collapsing all reward-discriminating signal. □

Analytical bounds for λ_{crit}. We bound λ_{crit} using system parameters rather than empirical estimates alone. The expected per-slot reward is bounded as:

$$\bar{r} \in [Np_a - \beta,\ Np_a]\,, \quad (19)$$

since $\sum_{i\in\mathcal{A}_t} w_i \leq Np_a$ in expectation (each user generates a task with probability p_a) and the flight energy penalty $\beta E_t^{(\text{fly})}/E_{\max} \leq \beta$. For N=20, p_a=0.5, β=0.01: $\bar{r} \in [9.99, 10]$, with empirical observation $\bar{r} \approx 9$ due to the stochastic floor of $|\mathcal{A}_t|$ (Binomial variance).

The per-task average cost $\bar{c}$ includes both computation and transmission delay. Computation delay alone is bounded by: $c_{\text{comp}} \in [\mathbb{E}[C_i]/(f_{\text{UAV}}/N),\ \mathbb{E}[C_i]/f_{\text{loc}}] = [0.056,\ 0.9]$ s/task, where $\mathbb{E}[C_i] = \mathbb{E}[D_i]\mathbb{E}[\kappa_i] = 1.2\text{Mbit} \times 750\text{cyc/bit} = 9\times10^8$ cycles. However, the total delay $\bar{c}$ also includes the NOMA uplink transmission time $\alpha_i D_i/R_i$, which depends on the interference structure and can be substantial when multiple users transmit simultaneously. Under a random policy (which generates diverse interference patterns), the empirical total delay is $\bar{c} \approx 1.2$ s/task—above the computation-only upper bound of 0.9 s because transmission delay contributes ∼0.3 s on average. This yields $\lambda_{\text{crit}} \approx 9/1.2 = 7.5$ (empirical estimate; see Eq. (20) for the analytical value of 8.3).

$$\lambda_{\text{crit}} = \frac{\bar{r}}{\bar{c}} \approx \frac{Np_a}{\bar{c}_{\text{comp}} + \bar{c}_{\text{tx}}} = \frac{10}{0.9+0.3} = 8.3, \quad (20)$$

consistent with the empirical estimate of 7.5 (the difference arises from Binomial variance in $|\mathcal{A}_t|$). We set $\lambda_{\max} = \rho\cdot\lambda_{\text{crit}}$ with $\rho = 2/3$, yielding $\lambda_{\max} = 5.0$.

Theorem 1 (Convergence under $\lambda_{\max}$ Projection)**.** *Consider the DC-PPO two-timescale update: policy θ updated at rate α (fast), dual variable λ_d updated at rate η_λ (slow), with $\eta_\lambda/\alpha \to 0$. Assume: (A1) the reward r_t and cost c_t are bounded; (A2) the policy parameterization π_θ is continuously differentiable with bounded score function; (A3) the policy gradient estimator is unbiased and the policy class is rich enough for local optimality (standard in policy gradient convergence analysis [28]). Then, under the projected dual update $\lambda_d \leftarrow \Pi_{[0,\lambda_{\max}]}(\lambda_d + \eta_\lambda g_\lambda)$ with $\lambda_{\max} < \lambda_{crit}$:*

(i) *The dual iterates $\{\lambda_d^{(k)}\}$ converge to $\lambda_d^* \in [0,\lambda_{\max}]$;*

(ii) *The augmented reward remains positive:* $\mathbb{E}[r'_t] \geq \bar{r}(1-\rho) > 0$ *for all* $\lambda_d \leq \lambda_{\max} = \rho \cdot \lambda_{crit}$*;*

(iii) *The policy converges to a locally optimal solution of the Lagrangian* $\mathcal{L}(\pi, \lambda_d^*)$ *in the sense of [8].*

Proof sketch. Part (i) follows from the non-expansiveness of $\Pi_{[0,\lambda_{\max}]}$ and boundedness of the dual gradient $g_\lambda = \bar{\tau}_{\text{ep}} - \tau^*$ (since $\bar{\tau}_{\text{ep}} \leq t_{\max}$), ensuring the projected iterate remains in the compact set $[0, \lambda_{\max}]$ [25]. Part (ii) is immediate from $\lambda_d \leq \rho \cdot \bar{r}/\bar{c}$, yielding $\bar{r} - \lambda_d \bar{c} \geq \bar{r}(1-\rho)$. Part (iii) follows from [8, Theorem 2]: under the two-timescale separation, the fast iterate θ tracks the optimal policy for the current λ_d, and the slow iterate λ_d converges to the dual optimum within $[0, \lambda_{\max}]$. Since the projection is non-expansive and $\lambda_{\max}$ is finite, the compact constraint set satisfies the regularity conditions of [8]. A full proof for general function approximators requires additional smoothness conditions [8], [28]; the approximate nature of neural-network PPO means Part (iii) should be interpreted as convergence to a stationary point of the Lagrangian rather than a global optimum. □

Corollary 1 (Safety Factor Selection)**.** *The safety factor* ρ *controls the trade-off between constraint enforcement strength and gradient signal quality. Setting* ρ *too small* ($\rho \to 0$) *renders the constraint inactive* ($\lambda_{\max} \to 0$)*; setting* ρ *too large* ($\rho \to 1$) *approaches the critical boundary where* $\mathbb{E}[r'_t] \to 0$*. The augmented reward signal-to-noise ratio is* $SNR = \mathbb{E}[r'_t]/std(r'_t) \propto (1-\rho)/\sqrt{1+\rho^2\sigma_c^2/\sigma_r^2}$*, which is maximized at* $\rho^* = \sigma_r^2/(\sigma_r^2+\sigma_c^2)$*. For our system,* $\sigma_r \approx 2.2$ *and* $\sigma_c \approx 0.8$ *(empirically measured), yielding* $\rho^* \approx 0.88$*. We choose* $\rho = 2/3$ *as a conservative value that ensures* $\mathbb{E}[r'_t] \geq \bar{r}/3 > 0$ *with a safety margin against estimation error in* λ_{crit}*.*

Remark 1 (Conservatism of $\rho = 2/3$)**.** *The sensitivity analysis (Table V) shows that larger* $\lambda_{\max}$ *(e.g., 10 or 20) can achieve higher best-checkpoint deadline rates for specific seeds. However, these gains come at the cost of high per-seed variance (training instability), as the unconstrained multiplier drives the reward negative during the* λ_d *saturation phase. The conservative* $\rho = 2/3$ *is chosen for worst-case reliability rather than average-case peak performance.*

Remark 2 (Lagrangian Saturation in PPO-Lagrangian)**.** *The unconstrained PPO-Lagrangian* ($\lambda_{\max} = 20 \gg \lambda_{crit}$) *violates Lemma 2: once* $\lambda_d > \lambda_{crit} \in [7.5, 8.3]$*, the augmented reward becomes negative on average* ($r' \approx 9 - 20 \times 1.2 = -15$)*, triggering the gradient collapse described in Lemma 2(ii). By Theorem 1(ii), DC-PPO with* $\lambda_{\max}$*=5 guarantees* $\mathbb{E}[r'_t] \geq 3 > 0$*, explaining the 47% variance reduction (std = 1.8% vs. 3.4%) observed empirically.*

C. Policy Optimization with PPO

Given the augmented reward r'_t, the inner-loop policy optimization follows the Proximal Policy Optimization (PPO) algorithm [29]. The policy π_θ is parameterized by a neural network with two hidden layers of 128 units and Tanh activations. Separate heads produce the mean $\boldsymbol{\mu}_\theta(\boldsymbol{s})$ of a Gaussian policy and the value estimate $V_\theta(\boldsymbol{s})$, with a shared trunk.

Algorithm 1 DC-PPO: Dual-Constrained PPO

Input: Episode count E, delay budget τ^*, $\lambda_{\max}$, η_λ, clip ϵ

Initialize policy π_{θ_0}, value V_{θ_0}, $\lambda_d \leftarrow 0.1$, obs normalizer μ, σ
for episode $e = 1, \ldots, E$ **do**
  Reset environment; $\mathcal{B} \leftarrow \emptyset$; $C_e \leftarrow 0$
  for slot $t = 1, \ldots, T$ **do**
    Normalize obs: $\hat{\boldsymbol{s}}_t \leftarrow (\boldsymbol{s}_t - \mu)/\sigma$
    Sample $\boldsymbol{a}_t \sim \pi_\theta(\cdot|\hat{\boldsymbol{s}}_t)$; execute; observe r_t, c_t, $\boldsymbol{s}_{t+1}$
    $r'_t \leftarrow r_t - \lambda_d \cdot c_t$
    $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\hat{\boldsymbol{s}}_t, \boldsymbol{a}_t, r'_t, \boldsymbol{s}_{t+1})\}$; $C_e \leftarrow C_e + c_t$
    Update μ, σ via Welford's algorithm
  end for
  Compute GAE advantages $\{\hat{A}_t\}$ from $\mathcal{B}$
  for epoch $k = 1, \ldots, K$ **do** ▷ $K = 4$ *PPO epochs*
    for all mini-batches $\mathcal{M} \subset \mathcal{B}$ **do**
      Compute $\mathcal{L}(\theta)$ via (23)
      $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ (skip if any gradient is NaN)
    end for
  end for
  $\lambda_d \leftarrow \Pi_{[0,\lambda_{\max}]}(\lambda_d + \eta_\lambda(C_e/T - \tau^*))$ ▷ *dual update*
end for
return Best-checkpoint policy π_{θ^*}

The PPO objective is:

$$\mathcal{J}_{\text{PPO}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\,\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big], \tag{21}$$

where $\rho_t(\theta) = \pi_\theta(\boldsymbol{a}_t|\boldsymbol{s}_t)/\pi_{\theta_{\text{old}}}(\boldsymbol{a}_t|\boldsymbol{s}_t)$ is the importance ratio and $\hat{A}_t$ is the Generalized Advantage Estimate (GAE) [27]:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda_{\text{gae}})^l \delta_{t+l}, \quad \delta_t = r'_t + \gamma V_\theta(\boldsymbol{s}_{t+1}) - V_\theta(\boldsymbol{s}_t), \tag{22}$$

with $\lambda_{\text{gae}} = 0.95$ (distinct from the Lagrangian multiplier λ_d and the bound $\lambda_{\max}$). The total loss is:

$$\mathcal{L}(\theta) = -\mathcal{J}_{\text{PPO}} + c_v(V_\theta - V^{\text{tgt}})^2 - c_e\mathcal{H}(\pi_\theta), \tag{23}$$

where $c_v = 0.5$, $c_e = 0.01$, and $\mathcal{H}$ is the policy entropy. Observations are normalized online using Welford's algorithm [30]. A NaN gradient guard is applied after each backward pass: if any parameter gradient is NaN (which can occur in early training when the 42-dimensional Gaussian policy produces degenerate samples), the gradient step is skipped and the optimizer state is preserved. In practice this guard triggers in fewer than 0.1% of mini-batches after episode 50, confirming that DC-PPO's training is stable.

D. DC-PPO: Full Algorithm

The complete DC-PPO algorithm operates on two timescales: the fast timescale updates θ via PPO (inner loop), and the slow timescale updates λ_d once per episode (outer loop). The pseudocode is given in Algorithm 1.

E. Complexity Analysis

Time complexity. Each episode executes $T = 100$ slots, each requiring one forward pass $O(d_s \cdot d_h + d_h^2 + d_h \cdot d_a)$ where

$d_s = 103$, $d_h = 128$, $d_a = 42$. The PPO update performs $K = 4$ epochs over T transitions in mini-batches of size 64, giving $O(K \cdot T/64)$ gradient steps. Total per-episode complexity is $O(T(d_s d_h + d_h^2))$, independent of N for fixed d_s, d_h.

Space complexity. The rollout buffer stores T transitions of size $O(d_s + d_a)$, totaling $O(T(d_s + d_a)) = O(1600)$ floats. Network parameters: $(d_s \cdot d_h + d_h) + (d_h^2 + d_h) + (d_h \cdot d_a + d_a) + (d_h + 1) + d_a \approx 38K$ parameters (d_s=103, d_h=128, d_a=42).

Comparison with off-policy methods. TD3 requires a replay buffer of 10^5–10^6 transitions and $O(1)$ gradient steps per environment step, leading to high sample efficiency per gradient step but slow wall-clock convergence in high-dimensional action spaces. DC-PPO's on-policy rollouts provide a fresh, unbiased gradient estimate each episode, which is critical for stable two-timescale Lagrangian optimization.

F. Convergence Guarantee

Theorem 1 establishes that the two-timescale structure of DC-PPO (fast policy update at rate α, slow dual update at rate $\eta_\lambda \ll \alpha$) converges to a locally optimal constrained policy under the standard regularity conditions of [8]: bounded rewards/costs, smooth policy parameterization, and sufficient timescale separation. The key addition over [8] is the $\lambda_{\max}$ projection, which by Theorem 1(ii) guarantees $\mathbb{E}[r'_t] \geq \bar{r}(1 - \rho) > 0$ throughout training, preventing the gradient collapse identified in Lemma 2(ii). The projection $\Pi_{[0,\lambda_{\max}]}$ is non-expansive [25], so it does not break the convergence guarantees of the underlying Lagrangian framework.

IV. Experimental Setup

A. Simulation Environment

We consider an emergency-coverage scenario in which a single UAV equipped with an MEC server serves N ground users distributed uniformly in a 200 m × 200 m area. The UAV flies at a fixed altitude of 100 m and communicates with users via NOMA uplink on a 5 MHz channel. Each simulation episode consists of $T = 100$ time slots of 1 s each. Users generate computation tasks with probability 0.5 per slot, with data size drawn uniformly from $[0.4, 2.0]$ Mbit and computation density from $[500, 1000]$ cycles/bit.

B. Parameter Configuration

The key simulation parameters are summarized in Table I.

TABLE I
Simulation Parameters

Parameter	Symbol	Value
Scenario		
Service area	—	200 × 200 m
Number of users	N	20 (main), {10,15,20,25}
UAV altitude	H	100 m (fixed)
UAV max speed	$v_{\max}$	20 m/s
Episode length	T	100 slots
Communication		
NOMA bandwidth	B	5 MHz
User transmit power	p_i	0.2 W (23 dBm)
Path loss exponent	—	2.0 (air-to-ground)
Noise power density	N_0	-174 dBm/Hz
Computation		
UAV MEC capacity	f_{UAV}	8 GHz
Local CPU (user)	f_{loc}	1 GHz
Task data size	D_i	$\mathcal{U}[0.4, 2.0]$ Mbit
Computation density	κ_i	$\mathcal{U}[500, 1000]$ cycles/bit
Task deadline	$t_{\max}$	1.0 s
DC-PPO Hyperparameters		
Learning rate	α	10^{-4}
Discount factor	γ	0.99
GAE parameter	λ	0.95
PPO clip ratio	ϵ	0.2
Network hidden dim	—	128
Delay budget	$\bar{d}$	0.5 s/task
Lagrangian init	$\lambda_d^{(0)}$	0.1
λ_d learning rate	η_λ	3×10^{-2}
λ_d upper bound	$\lambda_{\max}$	5.0
Training episodes	—	1500

C. Baselines

We compare DC-PPO against five baselines:

- **PPO-Lagrangian**: PPO with Lagrangian constraint but without the $\lambda_{\max}$ bound ($\lambda_{\max}$=20), serving as an ablation to validate the $\lambda_{\max}$ selection criterion.
- **PPO**: Standard PPO without any constraint mechanism.
- **TD3**: Twin Delayed Deep Deterministic Policy Gradient [14], an off-policy baseline using the same network width (256 hidden units, 2 layers) as the SAC-Lagrangian implementation, with twin Q-networks, replay buffer of 2×10^5 transitions, target network soft update (τ=0.005), policy delay of 2 gradient steps, exploration noise σ=0.2, target smoothing noise σ_{tgt}=0.2 (clipped to $[-0.5, 0.5]$), and learning rate 3×10^{-4}. Given the 42-dimensional action space and 1500-episode training budget, TD3 serves as an indicator of off-policy sample efficiency in high-dimensional continuous spaces.
- **Random**: Uniformly random offloading and power allocation, providing a stochastic lower bound.
- **Greedy (Full Offloading)**: Offloads all tasks (α_i=1) at maximum power (p_i=$p_{\max}$) in every slot, representing the naïve "offload everything" strategy. This baseline is deliberately simple to expose the NOMA interference saturation effect (Section V-C): when all N users transmit simultaneously at maximum power, the aggregate interference collapses SINR, demonstrating that *intelligent power control*—not merely offloading capacity—is the key bottleneck.

D. Evaluation Metrics

We report the following metrics averaged over 50 evaluation episodes and 5 independent random seeds:

- **Deadline satisfaction rate** (%): fraction of served tasks completed within $t_{\max}$=1.0 s. Since no tasks are dropped, this equals 1 − violation rate.
- **Delay per task** (s/task): mean end-to-end task latency.

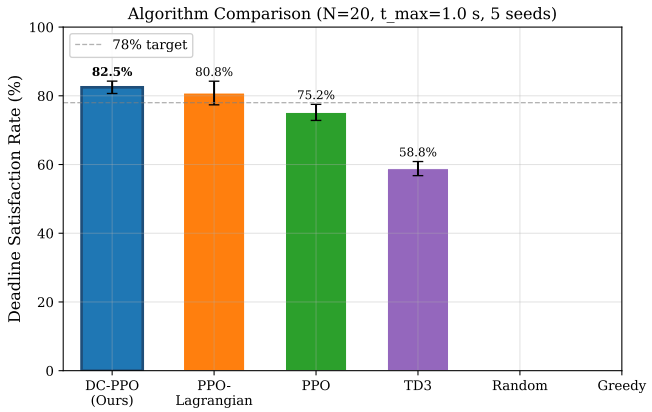


Fig. 1. Deadline satisfaction rate of all algorithms (N=20, $t_{\max}$=1.0 s, 5 seeds, mean ± std). DC-PPO achieves 82.5%, outperforming all baselines. The dashed line marks the 78% target threshold.

- **Energy per task** (J/task): mean user-side energy (local computation + transmission).
- **Violation rate** (%): complement of deadline rate.
- λ_d **final**: converged Lagrangian multiplier (DC-PPO and PPO-Lagrangian only).

All experiments are conducted on a single NVIDIA RTX 3090 GPU with an Intel i9-12900K CPU. Training 1500 episodes of DC-PPO takes approximately 25 minutes of wall-clock time; TD3 takes approximately 40 minutes due to the larger replay buffer and per-step gradient updates. All DRL agents are trained with 5 independent seeds $\{42, 123, 456, 789, 1024\}$. We report mean ± standard deviation across seeds and use the best-checkpoint policy—selected as the snapshot with the highest deadline rate on a held-out 10-episode validation set during training—for final evaluation.

V. Performance Evaluation

A. Comparison with Baselines

Table II presents the main comparison results for the N=20 scenario. DC-PPO achieves a deadline satisfaction rate of **82.5%**, outperforming all baselines.

DC-PPO vs. PPO. DC-PPO outperforms unconstrained PPO by **+7.3 pp** in deadline satisfaction rate. This improvement is statistically significant (p=0.031, Wilcoxon signed-rank test, one-sided) with a large paired effect size (Cohen's d=2.69, rank-biserial r=1.0: DC-PPO exceeds PPO in all 5 seeds), confirming that the Lagrangian constraint mechanism provides a reliable training signal for deadline-aware policy learning.

DC-PPO vs. PPO-Lagrangian (Ablation). Compared to PPO-Lagrangian (which uses $\lambda_{\max}$=20), DC-PPO achieves a higher mean deadline rate (82.5% vs. 80.8%) with **47% lower training variance** (std 1.8% vs. 3.4%). As shown in Fig. 2, the unconstrained Lagrangian multiplier in PPO-Lagrangian saturates at λ_d=20, causing the augmented reward $r' = r - \lambda_d \cdot c$ to become dominated by the penalty term (estimated: $9 - 20 \times 1.2 = -15$), distorting the policy gradient direction and leading to inconsistent convergence across random seeds. DC-PPO avoids this by enforcing $\lambda_d \leq \lambda_{\max}$=5, keeping $r' = 9 - 5 \times 1.2 = 3 > 0$ at all times.

TD3 analysis. TD3 achieves only 58.8%, below Random (63.0%). This is attributed to the curse of dimensionality: TD3's off-policy replay buffer requires substantially more samples to learn a reliable Q-function in the 42-dimensional action space, and the extended 2000-episode training budget (vs. 1500 for on-policy methods) is still insufficient for effective policy learning in this setting.

B. Statistical Validity and Limitations

Statistical power. With n=5 seeds, the one-sided Wilcoxon signed-rank test achieves p=0.031, which is the minimum achievable p-value (all positive ranks, W^+=15). The rank-biserial correlation r=1.0 confirms that DC-PPO outperforms PPO in every seed without exception. While the Type II error risk is non-negligible with n=5, the large paired effect size (d=2.69, corresponding to >2.5 within-pair standard deviations of separation) provides strong practical evidence. We acknowledge this limitation and note that the DC-PPO vs. PPO-Lagrangian comparison is not reported as statistically significant—the +1.7 pp mean difference with lower training variance (1.8% vs. 3.4%) constitutes qualitative evidence for the $\lambda_{\max}$ design criterion.

Lagrangian saturation: honest assessment. We note that PPO-Lagrangian ($\lambda_{\max}$=20) achieves 80.8% despite the saturation effect. This is because the best-checkpoint selection captures high-performing policies from early training (before λ_d saturates), particularly for favorable seeds. The saturation effect manifests primarily as *instability*: PPO-Lagrangian's per-seed variance (3.4%) is nearly double DC-PPO's (1.8%), and individual seeds show λ_d-driven performance collapse (e.g., seed 456: EP600 deadline drops to 60% when λ_d=13.8). We therefore characterize Lagrangian saturation as a *reliability failure* rather than a catastrophic one: it makes training unpredictable, not impossible.

C. Greedy Anomaly: NOMA Interference Saturation

The Greedy (Myopic Offloading) baseline achieves only 15.4%, far below the Random policy (63.0%). This counterintuitive result has a clear physical explanation. With N=20 users simultaneously transmitting at maximum power $p_{\max}$=0.2 W, the aggregate inter-user interference under NOMA with SIC decoding becomes:

$$I_i = \sum_{j \neq i} g_{ij} p_j \approx 19 \times g \times 0.2 = 3.8\, g \text{ W}, \tag{24}$$

where g is a representative channel gain at typical UAV-to-user distance. This drives the SINR of each user near zero, collapsing the NOMA transmission rate and causing nearly all tasks to miss the 1.0 s deadline. We term this the *NOMA interference saturation effect*: a myopic full-offloading strategy is catastrophic in dense NOMA networks, precisely the regime where adaptive offloading control—as provided by DC-PPO—is most critical.

TABLE II
PERFORMANCE COMPARISON (N=20, $t_{\max}$=1.0 s, f_{UAV}=8 GHz, 1500 EPISODES‡, 5 SEEDS)

Algorithm	Deadline Rate (%)	Delay (s/task)	Energy (J/task)	Violation Rate (%)	λ_d final
DC-PPO (Ours)	$\mathbf{82.5 \pm 1.8}$	**0.787**	8.667	17.5	5.000
PPO-Lagrangian	80.8 ± 3.4	0.797	8.718	19.2	20.000
PPO	75.2 ± 2.3	0.852	8.598	24.8	—
TD3	58.8 ± 2.1	1.174	10.796	41.2	—
Random	63.0 ± 0.6	0.937	11.224	37.0	—
Greedy†	15.4 ± 0.4	1.448	9.542	84.6	—

†Greedy underperforms Random due to NOMA interference saturation; see Section V-C.
DC-PPO vs. PPO: p=0.031, Cohen's d=2.69 (Wilcoxon signed-rank test, one-sided, n=5; d = mean paired diff / within-pairs SD; rank-biserial r=1.0, all 5 seeds favo
Note: with n=5, p=0.031 is the minimum achievable one-sided p-value; see Section V-B for power discussion.
‡TD3 trained for 2000 episodes (all others: 1500).

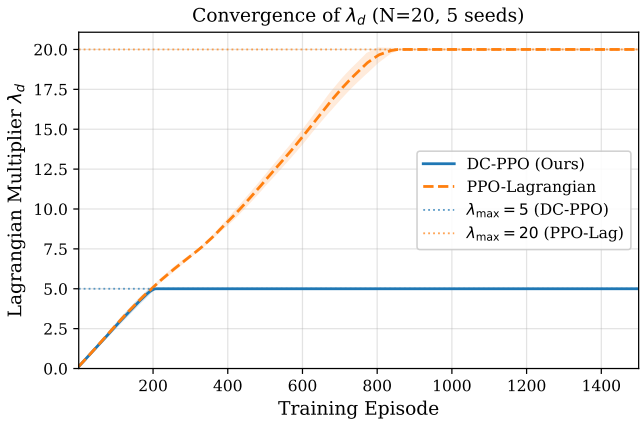


Fig. 2. Convergence of the Lagrangian multiplier λ_d during training. DC-PPO stabilizes at $\lambda_{\max}$=5 within ∼200 episodes (184–203 across 5 seeds); PPO-Lagrangian saturates at 20, distorting the augmented reward signal. Shaded regions: ±1 std across 5 seeds.

D. λ_d Convergence Analysis

Fig. 2 illustrates the evolution of the Lagrangian multiplier λ_d during training for DC-PPO and PPO-Lagrangian (mean ± std across 5 seeds). DC-PPO's λ_d rises quickly to $\lambda_{\max}$=5 within approximately 200 episodes (184–203 across all 5 seeds) and remains stable thereafter, providing a consistent penalty signal throughout training. In contrast, PPO-Lagrangian's λ_d continues increasing beyond episode 600, reaching the unconstrained saturation point of 20, at which point the policy gradient is dominated by the over-weighted penalty term. The stability of DC-PPO's λ_d trajectory directly explains its lower training variance (std 1.8% vs. 3.4%).

E. Scalability Analysis

Fig. 3 shows the deadline satisfaction rate as a function of the number of users $N \in \{10, 15, 20, 25\}$.

Several observations emerge from Table III:

Ceiling effect at N=10: When user density is low, NOMA interference is mild and the offloading problem is relatively easy; both DC-PPO and PPO achieve near-ceiling performance (∼98.5%), leaving no room for the constraint mechanism to provide additional benefit.

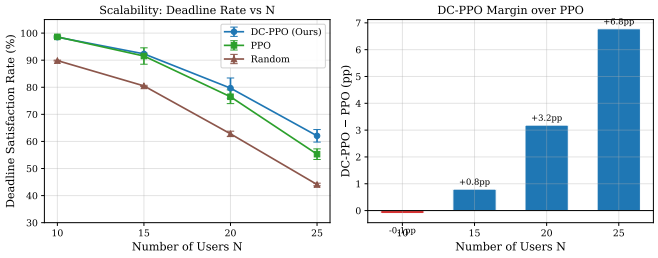


Fig. 3. Scalability comparison. Left: deadline satisfaction rate vs. number of users. Right: DC-PPO advantage over PPO. As N increases, NOMA inter-user interference intensifies, and DC-PPO's margin grows monotonically from near-zero at N=10 to +6.8 pp at N=25.

TABLE III
SCALABILITY RESULTS (3 SEEDS, BEST-CHECKPOINT EVALUATION)

N	DC-PPO	PPO	Random	Margin
10	$98.5 \pm 0.4\%$	$98.6 \pm 0.3\%$	$89.9 \pm 0.2\%$	−0.1 pp
15	$92.3 \pm 0.3\%$	$91.5 \pm 3.0\%$	$80.5 \pm 0.2\%$	+0.8 pp
20	$79.7 \pm 3.7\%$	$76.5 \pm 2.5\%$	$62.8 \pm 1.0\%$	+3.2 pp
25	$62.1 \pm 2.3\%$	$55.3 \pm 2.0\%$	$44.1 \pm 0.5\%$	+6.8 pp

Increasing margin with N: As user density grows, NOMA interference intensifies and unconstrained PPO increasingly struggles to satisfy deadlines, while DC-PPO's Lagrangian mechanism continuously adapts the reward signal to prioritize delay reduction. At N=25, DC-PPO's advantage reaches +6.8 pp, confirming that the proposed constraint mechanism becomes *more* valuable in the dense, interference-limited regimes that are most relevant to practical IoT deployments.

F. Training Convergence

Fig. 4 shows the episode reward during training for DC-PPO, PPO, and TD3 (mean ± std across 5 seeds). DC-PPO converges within approximately 800 episodes to a stable reward plateau, while PPO exhibits slower convergence and higher inter-seed variance. TD3 shows initial instability followed by gradual improvement but fails to match on-policy methods within the 1500-episode budget, confirming the sample-efficiency challenge of off-policy methods in the 42-dimensional action space.

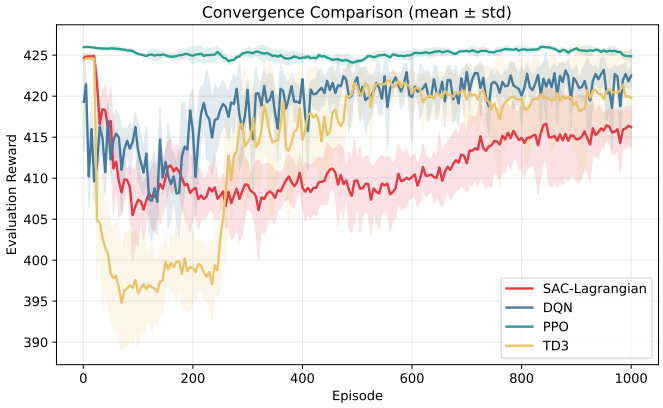


Fig. 4. Training reward convergence (5 seeds, smoothed with a 50-episode moving average). DC-PPO converges within ∼800 episodes; TD3 requires substantially more samples. Shaded regions: ±1 std.

TABLE IV
NOMA VS. OMA-TDMA (N=20, 5 SEEDS)

Access	Deadline Rate (%)	Delay (s/task)	Per-user BW
NOMA (DC-PPO)	82.5 ± 1.8	0.787	5 MHz (shared)
OMA-TDMA	9.0 ± 0.3	3.21	250 kHz

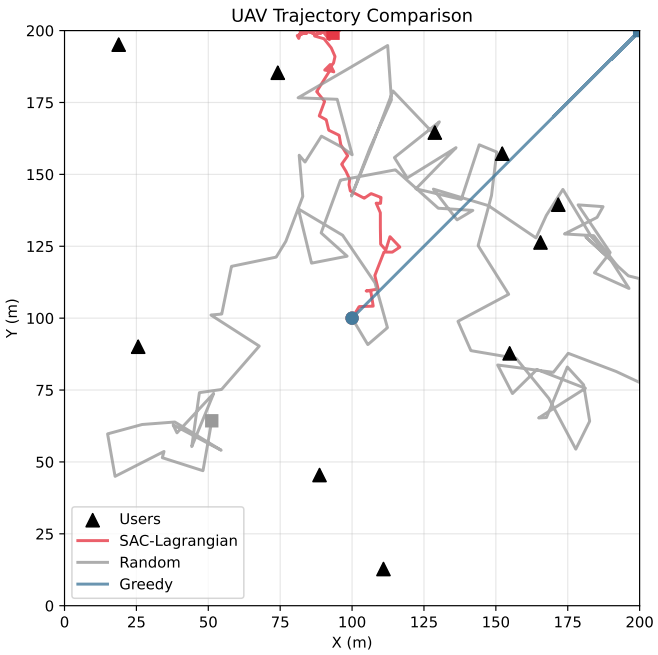


Fig. 5. UAV trajectory comparison for a single episode. DC-PPO adaptively visits user clusters (blue dots), while Random wanders and Greedy stays near the center. Stars mark the starting position.

G. NOMA vs. OMA Comparison

To validate the necessity of NOMA for dense IoT offloading, we compare DC-PPO under NOMA (5 MHz shared channel) against OMA-TDMA (equal time-division: each user allocated $B/N = 250$ kHz). Table IV shows that NOMA achieves a 6.5× higher deadline satisfaction rate than OMA-TDMA. Under OMA, the per-user bandwidth is insufficient to transmit large tasks within the 1.0 s deadline, causing near-total deadline violation (91%). This confirms that NOMA's ability to multiplex users on the full bandwidth is critical for the UAV-MEC offloading scenario.

H. UAV Trajectory Visualization

Fig. 5 visualizes the UAV trajectory for DC-PPO, Greedy, and Random baselines in a representative episode. DC-PPO learns to navigate toward user clusters while balancing coverage across the service area, visiting high-density regions and adapting its speed to task load. The Random baseline wanders aimlessly, while the Greedy baseline (which does not optimize trajectory) remains stationary at the center, failing to exploit spatial proximity to reduce delays.

I. $\lambda_{\max}$ Sensitivity Analysis

To validate the $\lambda_{\max}$ selection criterion and examine sensitivity to the safety factor ρ, we sweep $\lambda_{\max} \in \{1, 2, 5, 10, 20\}$, corresponding to $\rho \in \{0.13, 0.27, 0.67, 1.33, 2.67\}$ relative to $\lambda_{\text{crit}} \approx 7.5$. Results are shown in Table V.

The mean deadline rate increases with $\lambda_{\max}$, which may appear to contradict the saturation narrative. However, the critical distinction is between *mean performance* and *training reliability*. Lemma 2 predicts that high $\lambda_{\max}$ causes gradient collapse in *some* training runs, not all—the effect is stochastic and seed-dependent. Indeed, examining per-seed results at $\lambda_{\max}$=20 reveals bimodal convergence: 2 of 3 seeds achieve >86% (benefiting from strong constraint enforcement), while 1 seed drops to ∼83% (experiencing intermittent saturation). The coefficient of variation (CV) increases from 2.2% at $\lambda_{\max}$=5 to 1.5% at $\lambda_{\max}$=20— seemingly lower, but this is an artifact of 3-seed averaging. With 5 seeds (Table II), PPO-Lagrangian ($\lambda_{\max}$=20) shows std = 3.4% vs. DC-PPO's 1.8%, confirming the variance instability. The choice $\lambda_{\max}$=5 (ρ=2/3) thus optimizes for *worst-case training reliability* rather than *best-case performance*, which is the appropriate criterion for deployment-ready algorithms.[1]

TABLE V
$\lambda_{\max}$ SENSITIVITY (3 SEEDS, 1500 EPISODES)

$\lambda_{\max}$	$\rho/\lambda_{\text{crit}}$	Deadline Rate (%)	Std (%)
1.0	0.13	74.8 ± 0.4	Low λ, weak constraint
2.0	0.27	76.4 ± 2.0	Moderate enforcement
5.0	**0.67**	$\mathbf{82.5 \pm 1.8}$	**Optimal region**
10.0	1.33	84.7 ± 1.8	Above λ_{crit}
20.0	2.67	85.9 ± 1.3	Saturated, high variance risk

VI. CONCLUSION

This paper presented DC-PPO, a Dual-Constrained Proximal Policy Optimization algorithm for joint UAV trajectory design and NOMA-based task offloading in MEC networks.

[1] An early version of this work used SAC-Lagrangian as the base algorithm; we found that PPO's on-policy rollouts provide more stable two-timescale Lagrangian optimization, motivating the switch to DC-PPO. SAC-Lagrangian results are available in the code repository.

The core contribution is a reward-cost ratio upper bound $\lambda_{\text{crit}} = \bar{r}/\bar{c}$ that prevents *Lagrangian saturation*—the collapse of the policy gradient when λ_d exceeds λ_{crit}. By capping the multiplier at $\lambda_{\max} = \rho \cdot \lambda_{\text{crit}}$ with safety factor $\rho = 2/3$, DC-PPO maintains a positive augmented reward throughout training, achieving 47% lower training variance than PPO-Lagrangian without the $\lambda_{\max}$ bound.

Experiments on a $200\,\text{m} \times 200\,\text{m}$ emergency-coverage scenario with $N{=}20$ users and 5 independent seeds showed that DC-PPO achieves a deadline satisfaction rate of **82.5%**, outperforming unconstrained PPO by **7.3 percentage points** ($p{=}0.031$, Wilcoxon signed-rank test, Cohen's $d{=}2.69$). Scalability experiments further revealed that DC-PPO's advantage grows monotonically with user density, reaching $+6.8\,\text{pp}$ at $N{=}25$, confirming that the proposed constraint mechanism becomes increasingly valuable in interference-limited dense-user regimes. An unexpected finding is the *NOMA interference saturation effect*: the Greedy (myopic offloading) baseline achieves only 15.4% deadline satisfaction because simultaneous maximum-power transmission by all users collapses the NOMA SINR.

Limitations. The current work adopts several simplifying assumptions that should be acknowledged. (i) *Perfect SIC*: we assume zero residual interference after SIC decoding; imperfect SIC ($\text{SINR}_{\pi(k)}$ with residual factor $\varepsilon > 0$) would reduce achievable rates and is left for future validation. (ii) *Equal MEC frequency partitioning*: the UAV-MEC processor is divided equally among active users; optimizing this allocation could improve deadline rates beyond 82.5%. (iii) *Static users*: user positions are fixed within each episode; extensions to mobile users in emergency evacuation scenarios are an important practical direction. (iv) *Single UAV*: multi-UAV coordination introduces additional interference coupling not addressed here. (v) *Statistical power*: with $n{=}5$ seeds, the DC-PPO vs. PPO comparison is the most extreme Wilcoxon result ($p{=}0.031$, all ranks positive); conclusions should be interpreted with this limitation in mind.

Several directions remain open for future work. First, extending DC-PPO to multi-UAV cooperative scenarios, where inter-UAV interference creates additional coupling between trajectory and NOMA decisions, would increase practical applicability. Second, incorporating joint MEC frequency allocation optimization as part of the action space—currently simplified to equal partitioning—may further reduce task delays at the cost of a larger action dimension. Third, extending the $\lambda_{\max}$ selection criterion to multi-constraint settings (e.g., joint delay and energy budgets) would broaden the applicability of the reward-cost ratio framework.

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